

Get started with the certificate program

Welcome to the Google Advanced Data Analytics Certificate

Data professional

- A term used to describe any individual who works with data and/or has data skills

Machine learning

- An alternative approach to automation expressing the way you want a task down by using data instead of explicit instructions

Google Advanced Data Analytics Certificate overview

Hello, and welcome to the **Google Advanced Data Analytics Certificate**! The seven courses in this certificate program will advance your knowledge and help you grow in your career as a data analytics professional. A foundational background in data analytics is strongly encouraged.

Enter a growing field

Why are data analytics skills in such high demand? Companies report the largest skills gap in data analytics, and the demand for data analytics skills is growing over 15 times faster than the demand for the average skill in the US. As computing power has accelerated, industries have sought to leverage the increasing amount of data available to them. Companies are incorporating data analysis to better understand their customers and improve their products and services. Increasingly, companies are employing data professionals to help guide future business decisions. As a result, data professionals need the ability to better understand the challenges and possibilities of machine learning, build accurate predictive models, and clearly articulate their findings.

Throughout this program, you will explore concepts and scenarios that will help you grow your knowledge and skills as a data professional.

Google Advanced Data Analytics Certificate courses

The Google Advanced Data Analytics has seven courses. During each course, you will complete hands-on activities that are similar to work performed by data analytics professionals every day. In doing so, you'll gain experience that will help you effectively reveal the stories within the data and build models that can provide insight into trends. By the end of this program, you'll understand how to program using Python, prepare data for analysis, utilize statistics with data, build predictive models, optimize machine learning capabilities, and much more. The courses of the program are as follows:

1. **Foundations of Data Science**
2. **Get Started with Python**
3. **Go Beyond the Numbers: Translate Data into Insights**
4. **The Power of Statistics**
5. **Regression Analysis: Simplify Complex Data Relationships**
6. **The Nuts and Bolts of Machine Learning**
7. **Google Advanced Data Analytics Capstone**



Google Data Analytics Certificate content

These courses build on many foundational concepts examined in the [Google Data Analytics Certificate](#), and you will encounter links to content from the Data Analytics Certificate courses. This material is meant to provide an optional review and act as a useful resource.

Benefits for job seekers

After completing all seven courses, Google Advanced Data Analytics Certificate graduates get access to exclusive job search resources, courtesy of Google. You'll have the opportunity to:

- Build your resume, participate in mock interviews, and receive job search tips through Big Interview, a job-training platform that's free for program graduates.
- Improve your interview technique with Interview Warmup, a tool built by Google with certificate graduates in mind. Access data analytics-specific practice questions, transcripts of your responses, and automatic insights that help you grow your skills and confidence.
- Access thousands of job postings and free one-on-one career coaching with Career Circle. (You must be eligible to work in the U.S. to join.)
- Claim your Google Advanced Data Analytics Certificate badge, and share your achievement on LinkedIn to stand out among other candidates to potential employers.

Congratulations on taking this next step in your journey to a fulfilling career in data analytics!

Course 1 overview



Hello, and welcome to **Foundations of Data Science**, the first course in the Google Advanced Data Analytics Certificate. You're starting an exciting journey!

In this course, you will be introduced to the basic concepts of data science. You will discover how data professionals operate in the workplace and how different roles contribute to an organization's vision of the future. You will also begin to explore the data workflow PACE, which stands for plan, analyze, construct, and execute. The PACE workflow will help you better understand how to navigate the technical and workplace expectations of this career space.

Course descriptions

The Google Advanced Data Analytics Certificate has seven courses.

Foundations of Data Science is the first course.



1. **Foundations of Data Science** — *(current course)* Learn how data professionals operate in the workplace and how different roles in the field

of data science contribute to an organization's vision of the future. Then, explore data science roles, communication skills, and data ethics.

2. **Get Started with Python** — Discover how the programming language Python can power your data analysis. Learn core Python concepts, such as data types, functions, conditional statements, loops, and data structures.
3. **Go Beyond the Numbers: Translate Data into Insights** — Learn the fundamentals of data cleaning and visualizations and how to reveal the important stories that live within data.
4. **The Power of Statistics** — Explore descriptive and inferential statistics, basic probability and probability distributions, sampling, confidence intervals, and hypothesis testing.
5. **Regression Analysis: Simplify Complex Data Relationships** — Learn to model variable relationships, focusing on linear and logistic regression.
6. **The Nuts and Bolts of Machine Learning** — Learn unsupervised machine learning techniques and how to apply them to organizational data.
7. **Google Advanced Data Analytics Capstone** — Complete a hands-on project designed to demonstrate the skills and competencies you acquire in the program.

Course 1 content

Each course of this certificate program is broken into modules. You can complete courses at your own pace, but the module breakdowns are designed to help you finish the entire Google Advanced Data Analytics Certificate in about six months.

What's to come? Here's a quick overview of the skills you'll learn in each module of this course.



Module 1: Introduction of data science concepts

You will find out what's in store for Course 1 and the whole certificate program. You'll explore the Coursera platform, procedures, and content types, and meet other learners in the program. Then, you'll learn about data science fundamentals, and investigate the question "What is data science?" by examining elements of its history, and tracing its current defining

characteristics. Finally, you'll learn some tips for embarking on careers in this field.

Module 2: The impact of data today

In this part of the course, you will explore examples of the ways data-driven insights impact different industries. You'll learn how data analytics professionals are transforming organizations and discover how data professionals are approaching data stewardship. Then, you'll consider the trajectory of data analytics and forecast the future of data fields.

Module 3: Your career as a data professional

You will learn what it means to have a career as a data professional. In this section, you'll examine the required skills of a data analyst and compare it to many other data professional roles. Then you'll discover how data professionals operate within cross-functional teams and how they serve their organizations.

Module 4: Data applications and workflow

You will find out how to structure your work as a data analyst and learn about the PACE workflow, which will guide you as you learn how to think like a data professional. You'll also investigate what it means to be a successful communicator and how communication can help drive data analysis through the PACE workflow.

Module 5: Course 1 end-of-course project

You will work on a project that will help develop your resume and refine your ability to think like a data professional. You'll end the course by reflecting on your learning journey, and then you will get a preview of the next course.

What to expect

Each course offers many types of learning opportunities:

- **Videos** led by Google instructors teach new concepts, introduce the use of relevant tools, offer career support, and provide inspirational personal stories.
- **Readings** build on the topics discussed in the videos, introduce related concepts, share useful resources, and describe case studies.
- **Connect with other learners:** If you have a question, chances are, you're not alone. Visit Coursera's private [Advanced Data Analytics Community](#) to

expand your network, discuss career journeys, and share experiences. Check out the [quick start guide](#).

- **Self-review activities** and **labs** give you hands-on practice in applying the skills you are learning and allow you to assess your own work by comparing it to a completed example.
- **Interactive plug-ins** encourage you to practice specific tasks and help you integrate the knowledge you have gained in the course.
- **In-video quizzes** help you check your comprehension as you progress through each video.
- **Practice quizzes** allow you to check your understanding of key concepts and provide valuable feedback.
- **Graded quizzes** demonstrate your understanding of the main concepts of a course. You must score 80% or higher on each graded quiz to obtain a certificate, and you can take a graded quiz multiple times to achieve a passing score.

Tips for success

- It is strongly recommended that you go through the items in each lesson in the order they appear because new information and concepts build on previous knowledge.
- Participate in all learning opportunities to gain as much knowledge and experience as possible.
- If something is confusing, don't hesitate to replay a video, review a reading, or repeat a self-review activity.
- Use the additional resources that are referenced in this course. They are designed to support your learning. You can find all of these resources in the **[Resources](#)** tab.
- When you encounter useful links in this course, bookmark them so you can refer to the information later for study or review.
- Understand and follow the [Coursera Code of Conduct](#) to ensure that the learning community remains a welcoming, friendly, and supportive place for all members.

Introduction to Course 1

What you'll learn

- Approaches for larger-scale projects
- Trajectory of the field
- Effective communication
- Hands-on data analysis
- provide a few career tips and tactics

Helpful resources and tips

As a learner, you can choose to complete one or multiple courses in this program. However, to obtain the Google Advanced Data Analytics Certificate, you must complete all of the courses. This reading describes what is required to obtain a certificate and best practices for you to have a good learning experience on Coursera.

Obtain the Google Advanced Data Analytics Certificate

To receive your official Google Advanced Data Analytics Certificate, you must:

- Pass all graded assignments in all 7 courses of the certificate program. Each graded assignment is part of a cumulative graded score for the course, and the passing grade for each course is 80%.

AND **one** of the following:

- Pay the course certificate fee,
- Be approved for Coursera Financial Aid, or
- Complete the certificate through an educational institution, employer, or agency that's sponsoring your participation.

Healthy habits for course completion

Here is a list of best practices that will help you complete the courses in the program in a timely manner:

- **Plan your time:** Setting regular study times and following them each week can help you make learning a part of your routine. Use a calendar or timetable to create a schedule, and list what you plan to do each day in order to set achievable goals. Find a space that allows you to focus when you watch the videos, review the readings, and complete the activities.
- **Work at your own pace:** Everyone learns differently, so this program has been designed to let you work at your own pace. Although your personalized deadlines start when you enroll, feel free to progress through the program at the speed that works best for you. There is no penalty for late assignments; to earn your certificate, all you have to do is complete all of the work. You can extend your deadlines at any time by going to **Overview** in the navigation panel and selecting **Switch Sessions**. If you have already missed previous deadlines, select **Reset my deadlines** instead.
- **Be curious:** If you find an idea that gets you excited, act on it! Ask questions, search for more details online, explore the links that interest you, and take notes on your discoveries. The steps you take to support your learning along the way will advance your knowledge, create more opportunities in this high-growth field, and help you qualify for jobs.
- **Take notes:** Notes will help you remember important information in the future, especially as you're preparing to enter a new job field. In addition, taking notes is an effective way to make connections between topics and gain a better understanding of those topics.
- **Review exemplars:** Exemplars are completed assignments that fully meet an activity's criteria. Many activities in this program have exemplars for you to compare to your own work. Although there are often many ways to complete an assignment, exemplars offer you guidance and inspiration about how to complete the activity.
- **Build your career identity:** Your career identity is the unique value you bring to the workforce. [Watch this video](#) to learn about the key components of career identity and write your own career identity statement. Discovering and defining your own career identity makes you better equipped to choose a career path that aligns with your strengths, values, and goals and is more effective in your chosen profession.
- **Connect with other learners:** If you have a question, chances are, you're not alone. Visit Coursera's private [Google Advanced Data Analytics](#)

Community to expand your network, discuss career journeys, and share experiences. Check out the quick start guide.

- **Update your profile:** Consider updating your profile on Coursera to include your photo, career goals, and more. When other learners find you in the Community, they can click on your name to access your profile and get to know you better.

Documents, spreadsheets, presentations, and labs for course activities

To complete certain activities in the program, you will need to use digital documents, spreadsheets, presentations, and/or labs. Data analytics professionals use these software applications to collaborate within their teams and organizations. If you need more information about using a particular tool, refer to these resources:

- Microsoft Word: Help and learning: Microsoft Support page for Word
- Google Docs: Help Center page for Google Docs
- Microsoft Excel: Help and learning: Microsoft Support page for Excel
- Google Sheets: Help Center page for Google Sheets
- Microsoft PowerPoint: Help and learning: Microsoft Support page for PowerPoint
- How to use Google Slides: Help Center page for Google Slides
- Common problems with labs: Troubleshooting help for Qwiklabs activities

Module, course, and certificate glossaries

This program covers a lot of terms and concepts, some of which you may already know and some of which may be unfamiliar to you. To review terms and help you prepare for graded quizzes, refer to the following glossaries:

- **Module glossaries:** At the end of each module's content, you can review a glossary of terms from that module. Each module's glossary builds upon the terms from the previous modules in that course. The module glossaries are not downloadable; however, all of the terms and definitions are included in the course and certificate glossaries, which are downloadable.

- **Course glossaries:** At the end of each course, you can access and download a glossary that covers all of the terms in that course.
- **Certificate glossary:** The certificate glossary includes all of the terms in the entire certificate program and is a helpful resource that you can reference throughout the program or at any time in the future.

You can access and download the certificate glossaries and save them on your computer. You can always find the course glossary using the course's [Resources](#) tab. To access the **Advanced Data Analytics Certificate glossary**, click the following link and select *Use Template*.

- Link to the glossary: [Advanced Data Analytics Certificate glossary](#)

OR

- If you don't have a Google account, you can download the glossary directly from the following attachment.

[Advanced Data Analytics Certificate glossary](#)DOCX File

Data Analytics Certificate glossary

If you completed the original [Google Data Analytics Certificate](#), you may recognize some overlap with several of the glossary terms in this program. Refer to the Data Analytics Certificate glossary, linked in the [Resources](#) tab, to review these foundational terms and concepts. The definitions of some terms in the Data Analytics Certificate glossary differ from the definitions of the same terms in this program since the Advanced Data Analytics Certificate builds upon the concepts taught in the previous program.

Course feedback

Providing feedback on videos, readings, and other materials is easy. With the resource open in your browser, you can find the thumbs-up and thumbs-down symbols.

- Click **thumbs-up** for materials you find helpful.
- Click **thumbs-down** for materials that you do not find helpful.

If you want to flag a specific issue with an item, click the flag icon, select a category, and enter an explanation in the text box. This feedback goes back to the course development team and isn't visible to other learners. All feedback received helps to create even better certificate programs in the future.

For technical help, visit the [Learner Help Center](#)

Get the latest updates from Google

You're already taking control of your career path by enrolling in this course. Now, you can stay ahead of the curve by [subscribing to email updates](#) directly from Grow with Google.

You'll receive:

- **Curated career resources and tools.** Be among the first to know about new career-related resources from the Grow with Google team, and get tips about making the most of what we have to offer.
- **Alerts on the latest courses from Google:** Take advantage of the latest learning opportunities and develop new skills to stay competitive in a rapidly changing job market.
- **Pro tips to get the most out of AI.** Keep up to date and learn how to use AI to help make decisions, solve problems, and boost productivity and creativity.

Get the most of Google Career Certificates program by signing up [here](#)

This video discusses the importance of data analytics and introduces the Google Data Analytics Professional Certificate program.

Here's a breakdown:

Data Analytics in Today's World

- People generate massive amounts of data daily, creating a demand for professionals who can analyze it.
- Data professionals use their skills to organize this data and uncover meaningful insights.

Introducing the Google Data Analytics Professional Certificate

- This program equips learners with in-demand data analytics skills, potentially leading to job opportunities in 3-6 months.
- Learners will gain practical experience through hands-on activities, real-world examples, and a final capstone project.

Benefits of the Program

- The program is designed by Google experts and offers flexibility for learners to progress at their own pace.
- Upon completion, graduates gain access to a network of potential employers seeking skilled data professionals.